

Jira and Trello for DevOps

Complete Beginner to Professional Guide

 Audience: College Students, DevOps Beginners, & Software Engineering Interns

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What is DevOps?

Bridging Development and Operations for Continuous Delivery

- **Definition:** A cultural philosophy, set of practices, and software tools that integrate software development (Dev) and IT operations (Ops).
- **Core Objective:** Shorten the systems development life cycle and provide continuous delivery with high software quality.
- **The CALMS Framework:** Culture, Automation, Lean, Measurement, Sharing.
- **DevOps Lifecycle Loop:** Plan → Code → Build → Test → Release → Deploy → Operate → Monitor.

 **Key Insight:** DevOps is not just tools—it is a cultural shift powered by automated CI/CD pipelines and transparent project management.

Where Project Management Tools Fit in DevOps

Project Management as the Orchestration Control Center

1. Planning Stage

Capturing epics, user stories, features, and non-functional security requirements.

2. Tracking & Velocity

Monitoring work in progress (WIP), sprint capacity, and team burn-down.

3. Release Management

Tagging Git commits and deployment builds directly to user tickets.

4. Continuous Feedback

Routing production incidents and bugs back to engineering automatically.

 End-to-End Traceability: Every line of code, pull request, and build artifact maps back to a Jira/Trello issue.

Why DevOps Teams Need PM Tools

Eliminating Communication Chaos in Modern Software Engineering

✗ Problems Without PM Tools

- Siloed communication across messy Slack threads & emails.
- Lost bug reports & unassigned critical infrastructure tasks.
- Zero visibility into deployment readiness or release scope.
- Endless status meetings asking 'What is the status of task X?'
- No historical record of technical debt or architecture decisions.

✓ Benefits With Jira & Trello

- 100% transparency & clear accountability across Dev & Ops.
- Automatic linking of Git Pull Requests & CI/CD build statuses.
- Data-driven sprint velocity & capacity planning for leads.
- Streamlined Agile, Scrum, and Kanban workflow automation.
- Centralized repository for requirements, acceptance criteria, & logs.

Introduction to Jira Software

The Gold Standard for Enterprise Agile & DevOps Management

- **What is Jira?** Developed by Atlassian, Jira Software is the world's #1 issue and project tracking platform for software engineering teams.
- **Core Capabilities:** Scrum & Kanban boards, interactive roadmaps, customizable issue workflows, JQL search engine, and extensive app ecosystem.
- **Enterprise Trust:** Used by over 100,000+ top companies globally including Spotify, NASA, Cisco, Airbnb, and eBay.
- **DevOps Advantage:** Native deep integrations with GitHub, GitLab, Bitbucket, Jenkins, SonarQube, Docker, and AWS/Azure cloud ecosystems.
- **Scalability:** From small 5-person startup teams up to enterprise scaled agile frameworks (SAFe) with 10,000+ developers.

Jira Interface Tour

Navigating Key UI Elements for DevOps Workflows

The screenshot displays the Jira interface for a 'DevOps Team Project'. The top navigation bar includes 'Jira Software', 'Our work', 'Projects', 'Filters', 'Dashboards', 'People', 'Apps', a '+ Create' button, and a search bar. The left sidebar shows the project summary, 'Roadmap / Timeline', 'Backlog', 'Active Sprints', 'Releases & CI/CD', 'Reports & Analytics', and 'Project Settings'. The main area is titled 'DEVOPS Active Sprint Board' and shows four columns: 'TO DO (2)', 'IN PROGRESS (2)', 'IN REVIEW (PR) (1)', and 'DONE (2)'. Each column contains task cards with details like 'EPIC: Infra', task descriptions, IDs (e.g., DEV-101, DEV-102, DEV-103, DEV-104, DEV-105, DEV-100, DEV-099), and priority levels (HIGH, MEDIUM). Numbered callouts point to: 1. Jira Software, 2. Our work, 3. Backlog, 4. Active Sprints, 5. Roadmap / Timeline, 6. Search this board, and 7. + Create.

UI Element Guide

- **1 Navigation Bar:** Global access to projects, filters, & apps.
- **2 Projects Switcher:** Quick toggle between microservices.
- **3 Backlog View:** Grooming & prioritizing user stories.
- **4 Active Sprint Board:** Drag & drop Kanban/Scrum columns.
- **5 Timeline / Roadmap:** Gantt timeline for tracking Epics.
- **6 Quick Filters & JQL:** Filter by assignee, tag, or JQL query.
- **7 '+ Create' Button:** Quick modal to create Stories/Bugs.

Agile Frameworks in Jira: Scrum vs Kanban

Selecting the Right Methodology for Your DevOps Workflow



Scrum Framework (Structured)

- Fixed-length iterations called Sprints (1 to 4 weeks).
- Sprint Backlog commitment with fixed scope per sprint.
- Ideal for planned software feature development releases.
- Key Meetings: Sprint Planning, Daily Standup, Review, Retrospective.
- Metrics: Burndown chart & Velocity chart tracking story points.



Kanban Framework (Continuous)

- Continuous work flow without fixed-length sprint iterations.
- Work-in-Progress (WIP) limits set per column to prevent bottlenecks.
- Ideal for DevOps operations, incident response, & bug fixing.
- Flexible prioritization—new critical tasks can be added anytime.
- Metrics: Lead time, Cycle time, & Cumulative Flow Diagram.

Jira Issue Types & Hierarchy

Structuring Work Items from Strategy to Code Execution

 **Epic (Purple)** — Large body of work spanning multiple sprints (e.g., 'Migrate Auth Service to Kubernetes').

 **User Story (Green)** — Functional requirement from user perspective (e.g., 'As a user, I want OAuth2 login').

 **Task (Blue)** — Technical work item (e.g., 'Setup Terraform state bucket in AWS S3').

 **Bug (Red)** — Defect or failure in existing code (e.g., 'Fix memory leak in API container').

 **Sub-task (White)** — Sub-divided unit of work assigned to specific developers under a Story or Task.

Managing DevOps Backlog & Sprints in Jira

Planning, Estimating, and Executing Sprint Iterations

- **Backlog Grooming:** Ordering tasks by priority, refining acceptance criteria, and archiving stale issues.
- **Story Point Estimation:** Assigning effort relative scores using Fibonacci sequence (1, 2, 3, 5, 8, 13) or Planning Poker.
- **Sprint Planning Meeting:** Pulling prioritized stories into Sprint container based on historical team velocity.
- **Daily Standup & Tracking:** Developers updating ticket progress on the Active Sprint board daily.
- **Closing the Sprint:** Completing sprint, reviewing burndown metrics, and carrying over unfinished work.

💡 **Pro Tip for Interns:** Always estimate your tasks during planning and move your Jira cards before your team's daily standup meeting!

Jira Workflows & Automation for DevOps

Eliminating Manual Overhead with Intelligent Automated Triggers

- **Custom Workflows:** Define status pipelines matching your engineering steps: Open → In Progress → PR Open → QA Verified → Deployed.
- **Transition Rules & Enforcements:** Require Git PR links or peer review approvals before a ticket can transition to 'In Review'.
- **Jira Automation Engine:** Built-in no-code rule builder using WHEN / IF / THEN triggers.
- **DevOps Auto-Triggers Example 1:** WHEN a GitHub Pull Request is opened → THEN auto-transition Jira ticket to 'In Review'.
- **DevOps Auto-Triggers Example 2:** WHEN PR is merged & deployed to staging → THEN auto-transition ticket to 'Done' & alert Slack channel.

Jira CI/CD & Developer Tool Integrations

Connecting Code Commits, PRs, Builds, and Cloud Deployments

- **Smart Commits:** Include issue key in Git commits (e.g., `git commit -m 'DEV-104 Fix auth token expiry'`).
- **Development Sidebar Panel:** View associated branches, commits, PRs, and build status right inside the Jira ticket.
- **CI/CD Pipeline Integrations:** Supports GitHub Actions, GitLab CI, Bitbucket Pipelines, Jenkins, CircleCI, & ArgoCD.
- **Environment Deployment Tab:** Real-time visibility showing whether feature is in Development, Staging, or Production.



Integrated Pipeline Flow:

[Git Commit DEV-104] → [GitHub Actions Build Pass] → [Jira Status: In Review] → [AWS EKS Staging Deployed]

Advanced Jira: JQL, Dashboards & Metrics

Data-Driven Decision Making & Agile Performance Tracking

- **Jira Query Language (JQL):** SQL-like search syntax (e.g., project = DEVOPS AND status = 'In Progress' AND priority = High).
- **Custom Executive Dashboards:** Real-time widgets displaying sprint burndown, high priority bugs, & build failure alerts.
- **Sprint Burndown Chart:** Visualizes remaining story points vs time to predict sprint completion success.
- **Cumulative Flow Diagram (CFD):** Identifies workflow bottlenecks, queue sizes, and Work-In-Progress (WIP) accumulation.
- **Velocity Chart:** Tracks story points completed across historical sprints to guide team capacity planning.

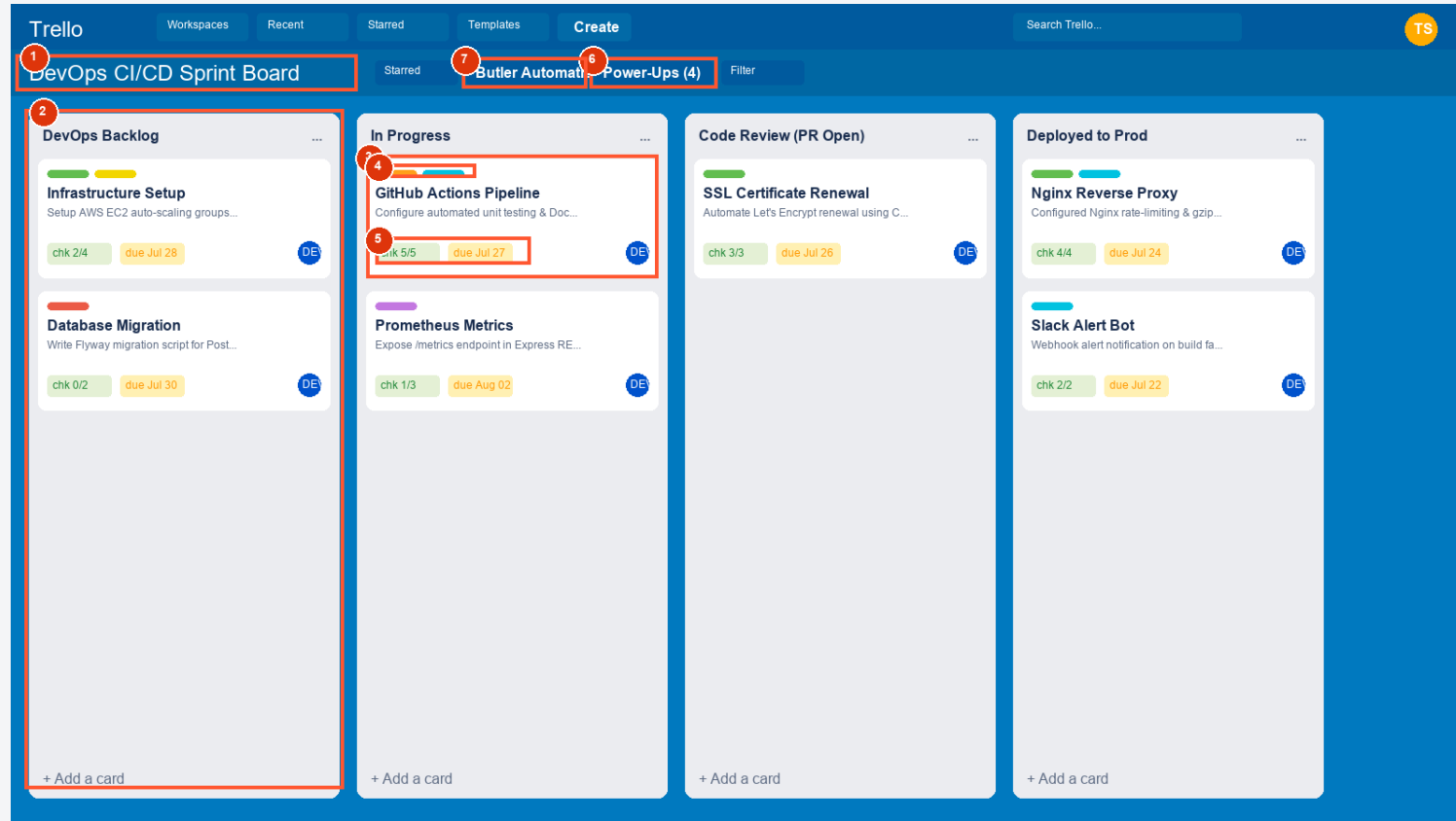
Introduction to Trello

Lightweight, Intuitive Kanban Visual Management

- **What is Trello?** A visual collaboration tool based on Kanban boards, also owned by Atlassian.
- **Core Structure:** Organized simply around Boards → Lists → Cards.
- **Key Strengths:** Zero learning curve, ultra-fast setup, highly visual card layout, & smooth drag-and-drop.
- **Ideal Use Cases:** Small DevOps teams, open-source projects, startups, personal task tracking, & lightweight bug boards.
- **Accessibility:** Fully responsive on Web, Mobile (iOS/Android), and Desktop apps.

Trello Interface Tour

Essential UI Components of a Trello Board



Trello UI Guide

- **1 Board Title & Workspace:** Header context & board settings.
- **2 Lists:** Vertical containers (Backlog, In Progress, Done).
- **3 Trello Cards:** Individual task units with description.
- **4 Color Labels:** Visual tags for categorizing task types.
- **5 Card Back Badges:** Checklist progress & due date alerts.
- **6 Power-Ups Menu:** Integrations with GitHub, Slack, etc.
- **7 Butler Automation:** Natural-language automation engine.

Trello Cards & Workflow Customization

Deep Dive into Card Features & Visual Customization

- **Card Front:** Concise title, color label bars, member avatars, due date badges, and checklist progress counters.
- **Card Back:** Detailed markdown description, file attachments, activity audit trail, & comments.
- **Interactive Checklists:** Sub-dividing task requirements with progress bars (e.g., 'Deployment Verification Checklist').
- **Custom Fields Power-Up:** Adding structured metadata fields like Risk Level, Environment, or Server Specs.
- **Card Covers & Colors:** Visual color coding for instant card recognition during standup reviews.

Trello Power-Ups & Butler Automation

Extending Trello into a Capable DevOps Command Hub

- **GitHub Power-Up:** Attach commits, pull requests, and branch build statuses directly onto Trello cards.
- **Slack Integration:** Send automated channel alerts whenever a card moves to 'Deployed to Prod'.
- **Butler No-Code Engine:** Built-in rule automation engine using natural language command triggers.
- **Example Butler Rule 1:** 'When a card is moved to Code Review, assign @lead-engineer and set due date in 48 hours.'
- **Example Butler Rule 2:** 'When a GitHub PR is merged, move the corresponding card to Done and add green label Deployed.'

Jira vs Trello: Feature Comparison Matrix

Side-by-Side Evaluation for DevOps Engineering Teams

Feature / Dimension	Jira Software	Trello
Target Audience	Mid-to-Large Enterprise Teams	Small Teams, Startups, Interns
Methodology	Full Scrum, Kanban, Scaled Agile (SAFe)	Flexible Kanban Board
Setup Complexity	High (Workflows, JQL, Permissions)	Extremely Low (Instant setup)
DevOps Integrations	Native deep CI/CD, Git, Deployment tabs	Power-Ups (GitHub, Slack)
Reporting & Metrics	Advanced (Burndown, CFD, Velocity)	Basic (via Power-Ups)
Best Fit For	Complex Microservice Pipelines	Lightweight Task & Bug Tracking

Real-World DevOps Workflow 1: End-to-End Jira

From Ticket Creation to Automated Production Release

1. Plan

Product Manager logs Story DEV-204 ('Automate DB Backup') in Jira Backlog.

2. Code

Developer creates Git branch feature/DEV-204 directly from Jira UI.

3. Build & Test

Developer opens GitHub PR; Jenkins triggers unit & integration tests.

4. Code Review

Jira Automation moves DEV-204 to 'In Review'; peer approves PR.

5. Deploy

PR merged into main; ArgoCD deploys Docker container to Kubernetes.

6. Release

Jira auto-transitions DEV-204 to 'Done' & updates Release Notes.

 **Key Takeaway:** Zero manual status dragging—Jira Automation syncs ticket state with Git & CI/CD events.

Real-World DevOps Workflow 2: Lightweight Trello

Fast Bug Tracking & Deployment Pipeline for Small Teams

1. Log Incident

Support creates Trello Card 'Bug: API 500 Error on Login' in Backlog.

2. Prioritize

Lead developer attaches Red Label 'Critical' and assigns backend dev.

3. Develop Hotfix

Card moved to 'In Progress'; GitHub Power-Up links hotfix commit.

4. CI/CD Build

GitHub Actions runs build; Butler adds checklist checkmark automatically.

5. Deploy & Alert

Card moved to 'Deployed'; Webhook sends instant notification to Slack.

DevOps Project Management Best Practices

Essential Guidelines for Students and Software Engineering Interns

DO's (Best Practices)

- ALWAYS include ticket key (e.g., DEV-101) in Git commit messages & PR titles.
- Break down large Epics into bite-sized user stories (< 3 days of work).
- Keep ticket statuses updated daily before morning standup meetings.
- Leverage automated workflow rules to reduce manual status dragging.
- Document deployment steps & rollback plans in ticket descriptions.

DONT's (Common Pitfalls)

- Don't leave abandoned or duplicate tickets cluttering the backlog.
- Don't over-complicate workflows with 15+ confusing transition states.
- Don't write code or open PRs without an open, assigned ticket.
- Don't ignore burndown charts or WIP limits when bottlenecks occur.
- Don't store sensitive secrets or credentials inside ticket comments.

Career Guidance & Skill Roadmap

Building In-Demand DevOps & Project Management Credentials

- **Resume Impact:** Highlight experience with Jira, Trello, Agile/Scrum, and CI/CD integrations on your CV.
- **Student Portfolio Projects:** Manage your university capstone or open-source group projects using a public Jira/Trello board.
- **Atlassian Certifications:** Atlassian Certified Associate (ACA) & Atlassian Certified Professional (ACP).
- **Agile Certifications:** Certified ScrumMaster (CSM) or Professional Scrum Master (PSM I).
- **Key Career Differentiator:** Bridging the gap between software engineering code and business project visibility.

Summary & Next Steps

Key Takeaways & Opening Floor for Q&A

1. Control Center

PM tools like Jira & Trello are the orchestration layer of DevOps.

2. Jira Power

Enterprise depth, JQL, metrics, & CI/CD deployment tracking.

3. Trello Agility

Fast visual Kanban management with Power-Ups & Butler.

4. Traceability

Link every commit, PR, & build artifact to project tickets.



Thank You for Participating!
Questions & Answers Session — Open Floor